

Mithilesh Vaidya

vaidya.mithilesh@gmail.com ▪ methi1999.github.io ▪ LinkedIn

Education

- **Georgia Institute of Technology**, Atlanta, USA [2022 - 2024]
 - Master's in Computer Science (Specialisation: Machine Learning)
- **Indian Institute of Technology Bombay**, Mumbai, India [2017 - 2022]
 - B.Tech+M.Tech in Electrical Engineering (**CPI 9.51/10**) with a Minor in Computer Science and Engineering

Professional Experience

- **Audio/Video Generation and Understanding** June'24 - Present
Descript, San Francisco AI Researcher
 - Co-led **Audio Regenerate**, a neural-codec + flow-matching model for word-level speech inpainting in Descript
 - Co-authored **PoDAR**, a power-disentangled audio representation that speeds generator convergence by **2x**
 - Built a video pipeline for **Video Regenerate**, an audio-driven lip-sync model for editing and translation
 - Improved a contrastive model for correcting frame-level audio-visual misalignment
 - Developing a modern audio-text forced aligner for word-level timestamps
- **LLM-powered Natural Language Interface** May'23 - Aug'23
Qualcomm, San Diego | Guide: Mr. Vasudev Nayak, Principal Engineer Machine Learning Internship
 - Designed a hierarchical LLM pipeline (MPT-7B) for structured parsing, with methods to reduce hallucinations
- **Verification of FPGA-based High Frequency Trading Platform** Apr'20 - June'20
APT Portfolio Pvt. Ltd. | Guide: Mr. Vivek Pannikar Hardware Engineering Internship
 - Implemented SystemVerilog-C DPI for **3x** faster verification, auto-generating JSON bindings via Python metaclasses
- **Autonise AI** Sep'18 - May'19
Machine Learning Startup Co-Founder
 - Built a UNet and PixelLink+GRU detector (**74%** accuracy) for document field extraction, served on AWS

Publications and Preprints

- M. Bendel, S. W. Bailey, **M. Vaidya** et al. "Goodbye Drift: Anchored Tree Sampling for Long-Horizon V2V Generation." [arXiv:2605.20476](#) (2026).
- A. Luebs, **M. Vaidya** et al. "PoDAR: Power-Disentangled Audio Representation." [arXiv:2605.10084](#) (2026).
- **M. Vaidya**, B. K. Sahoo and P. Rao. "Deep Learning for Assessment of Oral Reading Fluency." [arXiv:2405.19426](#) (2024).
- **M. Vaidya**, K. Sabu and P. Rao. "Deep Learning for Prominence Detection In Children's Read Speech." [ICASSP](#) (2022).
- K. Sabu, **M. Vaidya** and P. Rao. "CNN Encoding of Acoustic Parameters for Prominence." [arXiv:2104.05488](#) (2021).

Projects

- **Switching Subspace Model for Neural Population Analysis** Sep'22 - May'24
Guide: Prof. Anqi Wu, GeorgiaTech | [PDF](#) Research project
 - Developed a sequential VAE for multiple latent dynamics, with a unimodal Gaussian Process prior
 - Added CNN supervision to preserve behavioral variables of interest
- **Prominence Detection and Oral Reading Fluency of Children's Read Speech** June'21 - June'22
Guide: Prof. Preeti Rao, IIT Bombay | [ICASSP](#), [arXiv:2405.19426](#) Master's Thesis
 - Replaced an RFC baseline with a **CRNN** and Sinc convolution for word-level prominence, using POS tags and BERT
 - Built a **Wav2vec2.0** end-to-end fluency model that outperformed a hand-crafted RFC baseline by **0.06** Pearson

Honors and Awards

- **Institute Silver Medal** at IIT Bombay for best academic standing in department (**2/70**) [2022]
- **Undergraduate Research Award** (URA03) at IIT Bombay for outstanding thesis contributions [2022]
- **Institute Special Mention** for Journalism carried out as part of Insight, IIT Bombay's student media body [2022]